Supporting information

Paying colonization credit with forest management could accelerate northward range shift of temperate trees

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1 Supporting figures

1.1 Figure S1

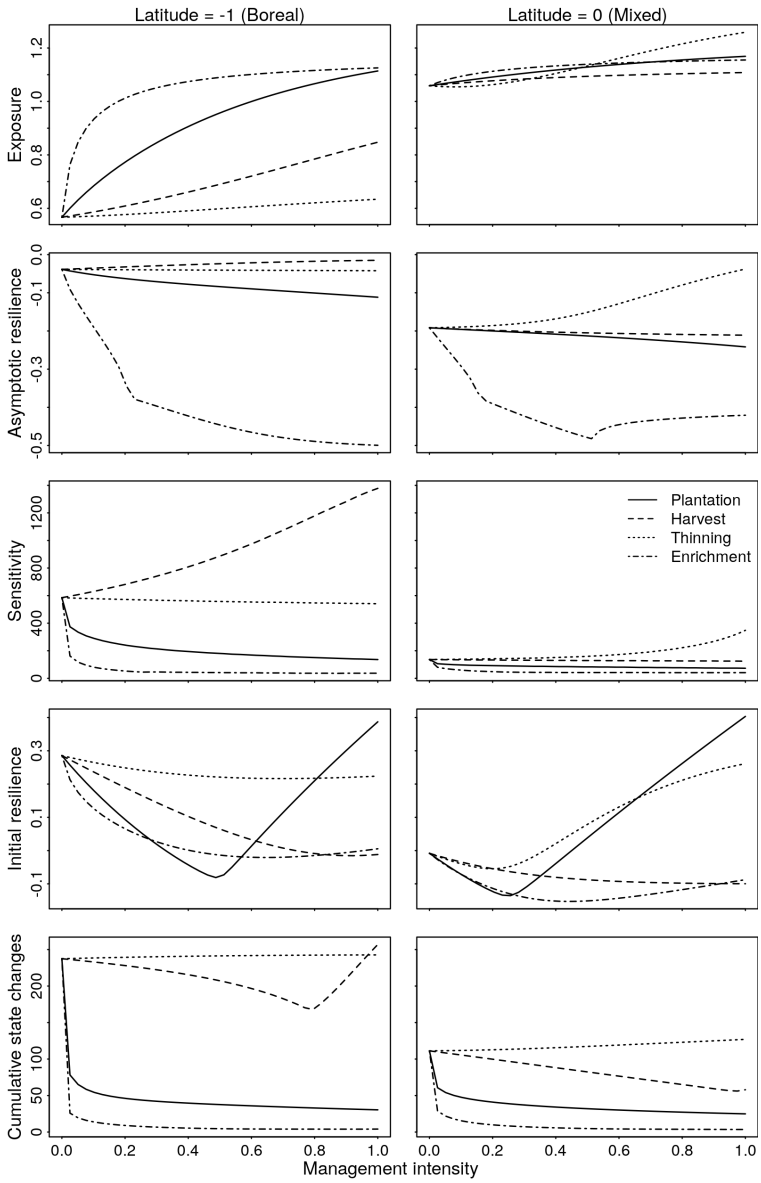

Figure 1: Transient dynamics following warming temperature along with the increasing management intensity for plantation, harvest, thinning and enrichment planting. Climatic condition is fixed at the boreal (annual mean temperature of -1; left panels) and the boreal/mixed transition (annual mean temperature of 0; right panels) regions. Transient dynamics are described by (i) exposure or the shift of forest states to the new equilibrium; (ii) asymptotic resilience or the rate in which the system recovery to equilibrium; (iii) sensitivity or the time for the state reach equilibrium after warming temperature; (iv) initial resilience or the reactivity of the system after warming temperature; and (v) vulnerability or the cumulative amount of state changes after warming temperature.

1.2 Figure S2

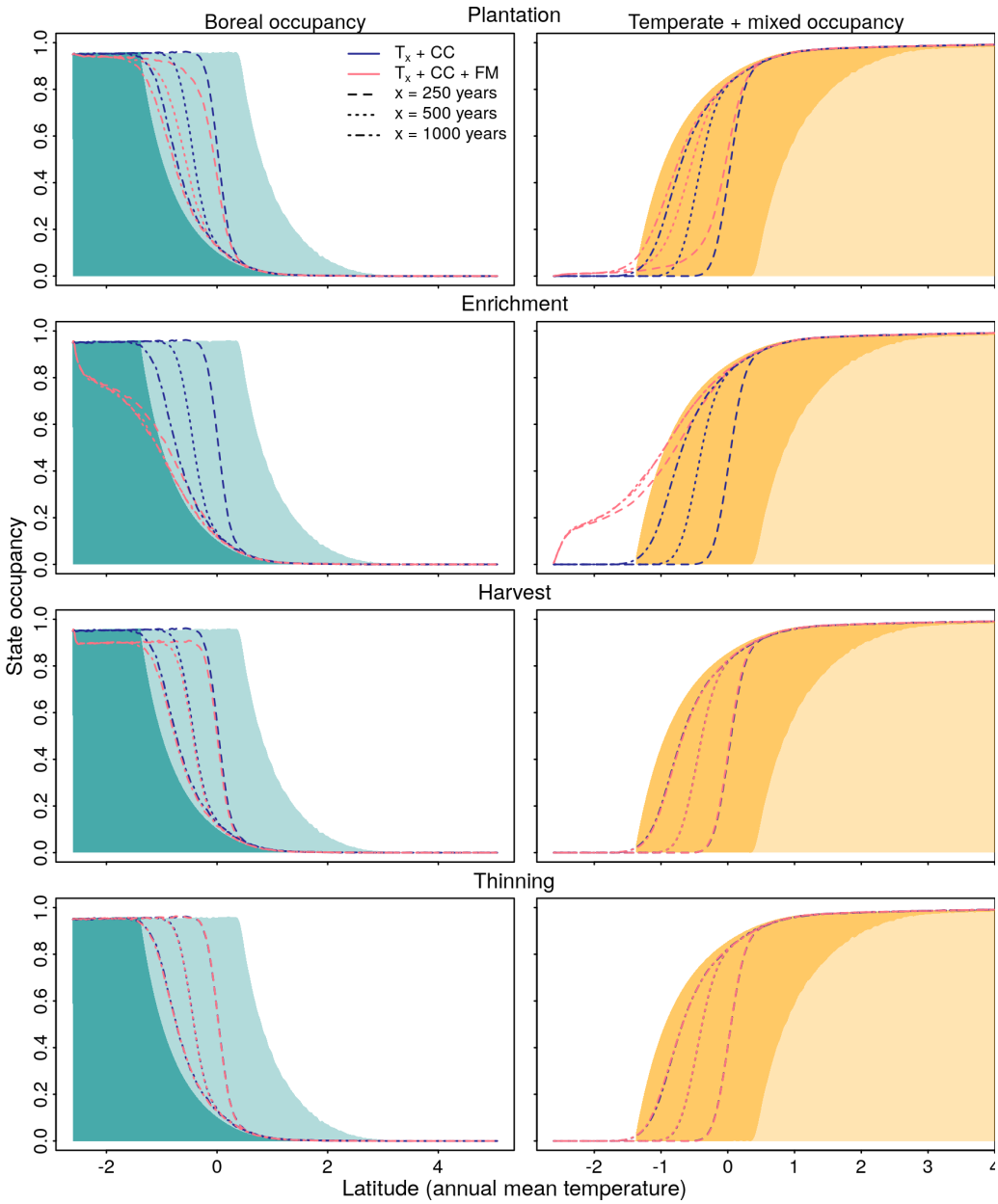

Figure 2: Boreal (left panels) and temperate plus mixed (right panels) occupancy along the latitudinal gradient of the boreal-temperate ecotone. Light and dark shaded areas are a reference of the state occupancy at equilibrium before and after warming temperature, respectively. Each line is a different simulation to differentiate the isolated and interacting effects of climate change (CC) and forest management (FM). The results are the mean of 15 replications. Increasing simulation time to 250, 500 and 1000 years.

1.3 Figure S3

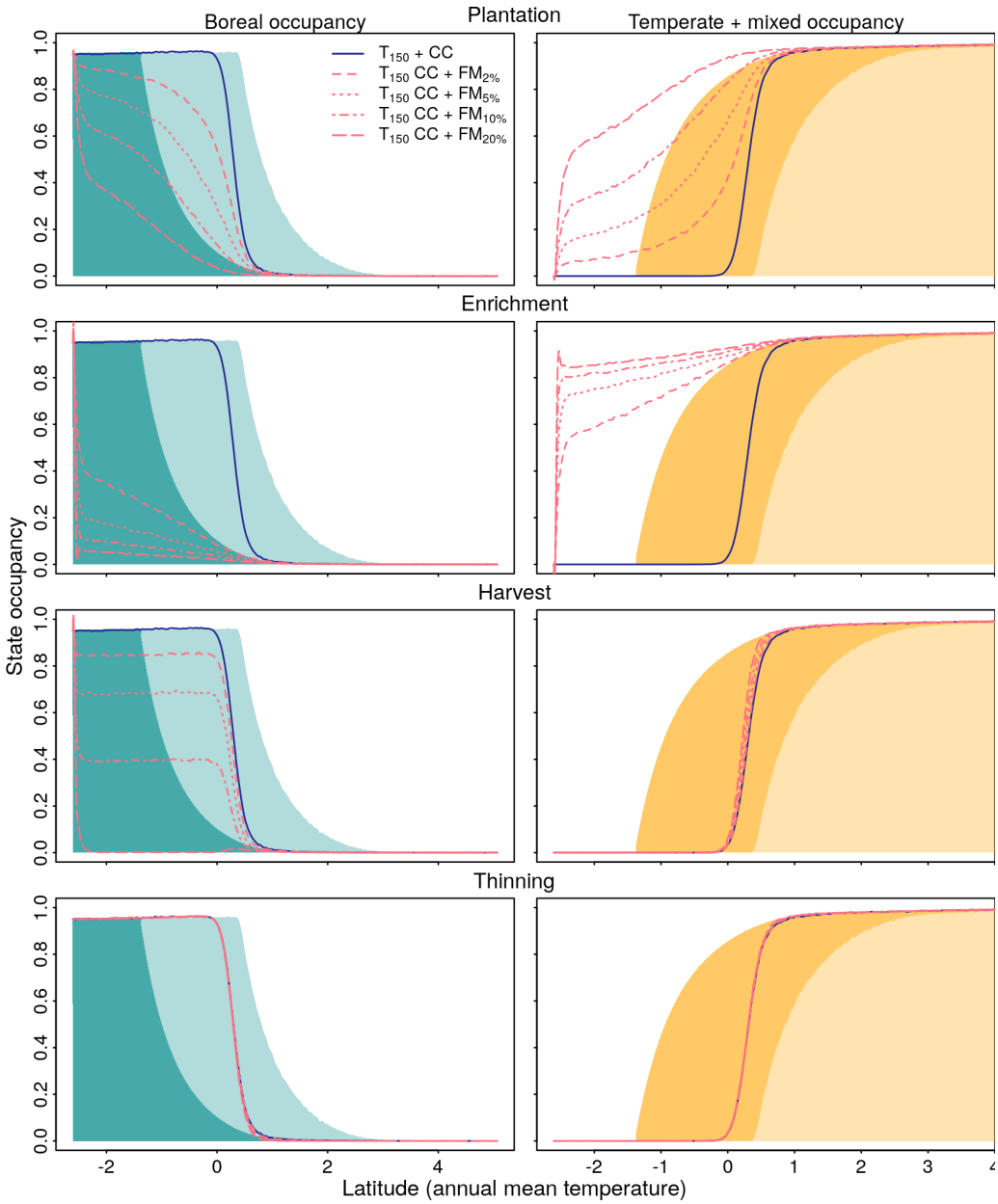

Figure 3: Boreal (left panels) and temperate plus mixed (right panels) occupancy along the latitudinal gradient of the boreal-temperate ecotone. Light and dark shaded areas are a reference of the state occupancy at equilibrium before and after warming temperature, respectively. Each line is a different simulation to differentiate the isolated and interacting effects of climate change (CC) and forest management (FM). The results are the mean of 15 replications. Increasing management intensity in 2, 5, 10 and 20%.

1.4 Figure S4

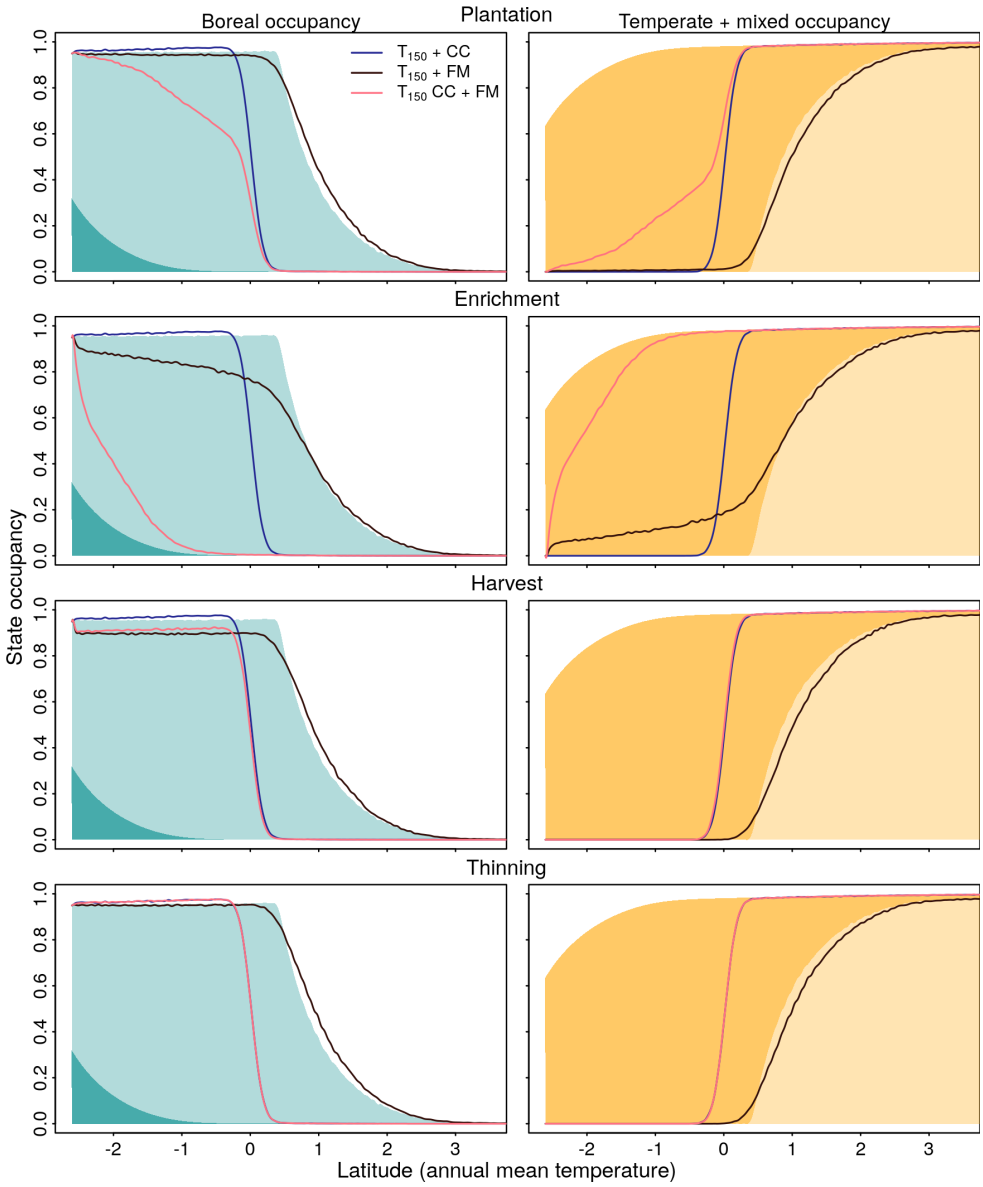

Figure 4: Boreal (left panels) and temperate plus mixed (right panels) occupancy along the latitudinal gradient of the boreal-temperate ecotone. Light and dark shaded areas are a reference of the state occupancy at equilibrium before and after warming temperature, respectively. Each line is a different simulation to differentiate the isolated and interacting effects of climate change (CC) and forest management (FM). The results are the mean of 15 replications. For the sake of simplicity, the confidence interval is omitted as we found little variation between replicates. Management intensity was set to 0.25% for plantation, thinning and enrichment planting, and 1% for harvest. The climate change scenario was RCP 8.5.

1.5 Figure S5

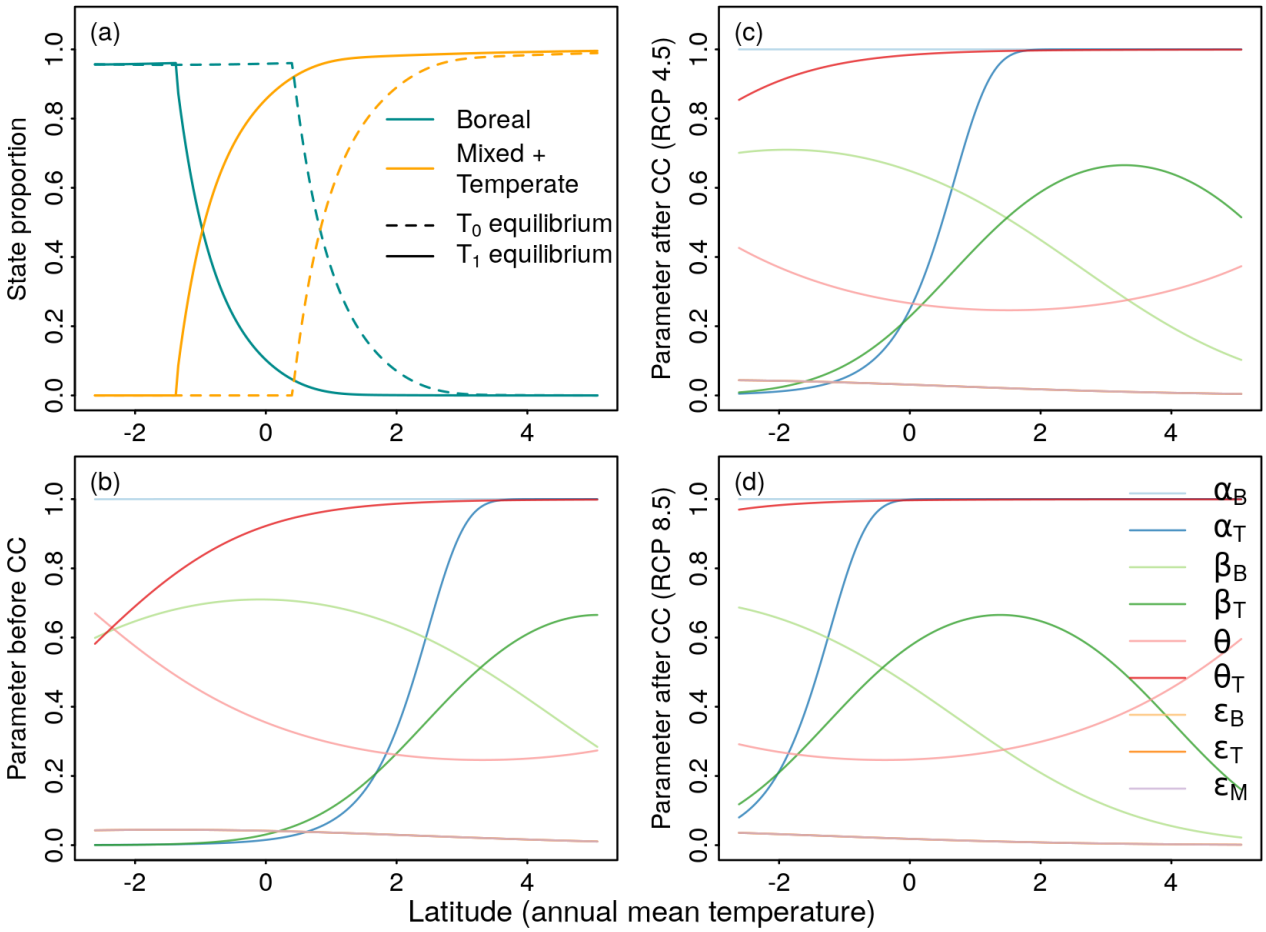

Figure 5: Parameters of the State and Transition Model varying as a function of annual mean temperature from Vissault et al. (2020). Annual mean precipitation is fixed to 998.7 mm. Parameters for (b) before and after warming temperature following (c) RCP4.5 and (d) RCP8.5 climate change scenarios over the same latitudinal position.

References

Vissault, Steve, Matthew V. Talluto, Isabelle Boulangeat, and Dominique Gravel. 2020. "Slow demography and limited dispersal constrain the expansion of north-eastern temperate forests under climate change." *Journal of Biogeography (Forthcoming)*.